

The geology and Mesozoic collisional history of the Cordillera Real, Ecuador

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ABSTRACT

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The geology of the metamorphic rocks of the Cordillera Real of Ecuador is described in terms of five informal lithotectonic divisions. We deduce that during the Mesozoic repeated accretionary events occurred and that dextral transpression has been of fundamental importance in determining the tectonic evolution of this part of the Northern Andes.

The oldest event recognised, of probable Late Triassic age, may be related to the break-up of western Gondwana and generated a regional belt of 'S-type' plutons. During the Jurassic, major calc-alkaline batholiths were intruded. Following this, in latest Jurassic to Early Cretaceous time, a volcano-sedimentary terrane, of possible oceanic or marginal basin origin (the Alao division), and the most westerly, gneissic Chaucha–Arenillas terrane, were accreted to continental South America. The accretion of the oceanic Western Cordillera took place in latest Cretaceous to earliest Tertiary time. This latter event coincided with widespread thermal disturbance, as evidenced by the large number of young K–Ar mineral ages recorded from the Cordillera Real.

Introduction

Important early contributions to the knowledge of the geology of the metamorphic rocks of the Ecuadorian Cordillera Real were made by Wolf (1892) and later by Sauer (1958, 1965) and, in the sub-Andean zone, by Tschopp (1953). Reconnaissance mapping over the Cordillera began in the late 1960's and, in spite of the problems of access and inhospitable climate, several 1:100,000 map sheets have been published and others are in the process of being surveyed. The results of this work, much of which was carried out by Kennerley (1971, 1973, 1980), Bristow (1973), Bristow and Guevara (1980) and Bristow et al. (1975) (see also Feininger, 1975, 1982; Trouw, 1976; Herbert,

1983) has been summarised by Baldock (1982) and incorporated into the 1:1,000,000 scale national map. However, in spite of the considerable efforts of the geologists concerned, large tracts of the metamorphic rocks within the Cordillera Real remained undifferentiated.

The current, ongoing study, a bilateral Technical Cooperation Project between the governments of Ecuador (Instituto Ecuatoriano de Minería—INEMIN) and the United Kingdom (Overseas Development Administration—ODA), began in 1986 and more than 20 traverses across the Cordillera have now been completed. The following account summarises the results of part of this work and, in particular, describes the geology of the Cordillera in terms of a series of informal, lithotectonic divisions. Although a number of fundamental questions remain unanswered, a preliminary evolutionary model, which deals essentially with the Mesozoic history of the Cordillera Real, is presented.

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Regional setting

The Ecuadorian Andes make up the southern portion of the north-northeast-trending Northern Andes (Gansser, 1973) and comprise two distinct cordilleras. The basement of the Western Cordillera and the coastal plain is considered to consist of an allochthonous slab of Cretaceous (post-Aptian/Albian) oceanic crust which was accreted onto the South American continent along

the line of the Calacali–Pallatanga–Palenque fault (Fig. 1), during latest Cretaceous to Early Tertiary time (Lebrat et al., 1985, 1986; Aspdén et al., 1987a, 1988). Its history and subsequent development have been recently dealt with by Eguez (1986), Megard (1987), Daly (1989) and Van Thournout and Quevedo (1990).

Immediately to the east of the Western Cordillera lies the narrow inter-Andean graben (Fig. 1), a more or less continuous topographic

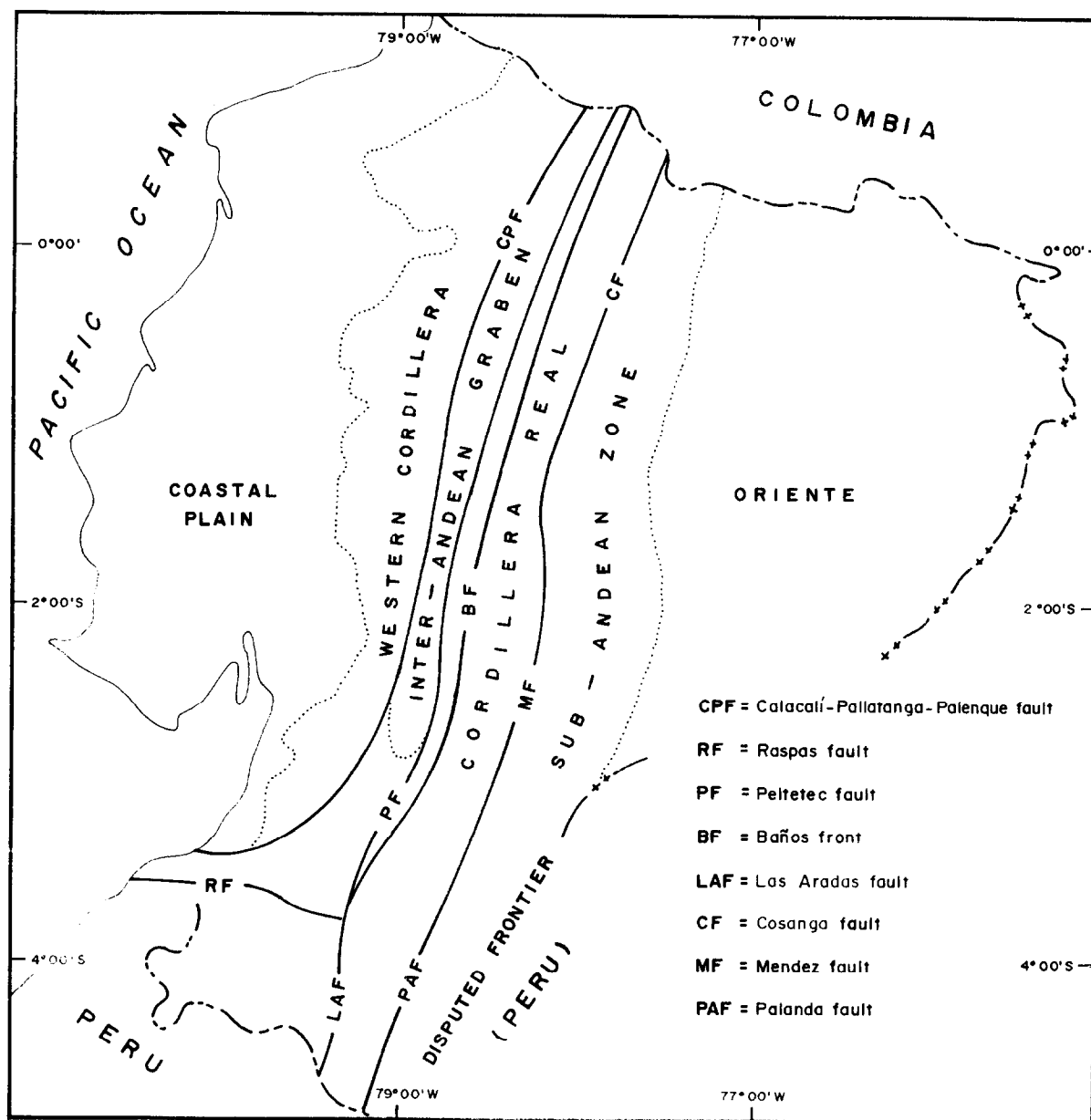


Fig. 1. Principal faults and geomorphological features of Ecuador.

depression which, although largely covered by Plio–Pleistocene volcanic deposits, can be traced from Colombia in the north as far south as c. 3°S. The structural limits of the graben, which separates the Western Cordillera from the Eastern Cordillera (Cordillera Real), are defined by the Calacali–Pallatanga–Palenque fault in the west and by the Peltetec fault in the east (Fig. 1). It has been suggested that these two faults represent crustal sutures and that sandwiched between them is a narrow wedge of allochthonous material (the Chaucha–Arenillas terrane), the southern limit of which is the Raspas fault (Fig. 1) (Aspden et al., 1988). In the south the terrane is well-exposed and comprises granitic gneiss, cordierite gneiss, amphibolite, schist, phyllite and quartzite. To the north, however, it is largely buried by younger volcanic deposits but the discovery of inliers of mica $\pm$ andalusite $\pm$ sillimanite schist, quartzo-feldspathic $\pm$ andalusite gneiss and amphibolites to the west of Cuenca (INEMIN—Misión Belga, 1986) and the occurrence of cordierite gneiss xenoliths in the Pichincha volcano (Bruet, 1987) immediately to the west of Quito, led Aspden et al. (1988) to suggest that this terrane could in fact floor much of the Ecuadorian inter-Andean graben.

South of 3°S, both the Western Cordillera and the inter-Andean graben disappear and are replaced by the east–west-striking, allochthonous, metamorphic rocks of the El Oro province of southwest Ecuador (Feininger, 1987; Aspden et al., 1988; Mourier et al., 1988). The Cordillera Real, however, continues southwards into Peru as a marked topographic feature, the western margin of which coincides with the Las Aradas fault (Kennerley, 1973) (Fig. 1). The Las Aradas fault can itself be traced northwards into the Baños front, a structure of regional importance, the nature of which is discussed later. The eastern limit of the Cordillera Real corresponds to a series of relatively high-angle, westerly-dipping thrusts, the Cosanga, Mendez and Palanda faults (Fig. 1), that bring into tectonic contact Cordilleran metamorphic rocks with essentially unmetamorphosed Cretaceous sedimentary rocks and a regional belt of undeformed, Jurassic, plu-

tonic and volcanic rocks in the sub-Andean zone (Figs. 1 and 2).

Pre-Cainozoic geology of the Cordillera Real

As a result of earlier work carried out in the Cordillera Real (see Baldock, 1982) some areas had been previously given a formalised, stratigraphic nomenclature. However, with few exceptions, these units proved to be unworkable on a regional scale, and were therefore abandoned and replaced by a more flexible system of informal, lithotectonic divisions and subdivisions. Five main lithotectonic divisions are presently recognised: the Guamote, Alao, Loja, Salado and Zamora divisions. The salient features of these are described below and summarised in Table 1.

Guamote division

The Guamote division crops out as a series of inliers located along the western flank of the central sector of the Cordillera Real between Riobamba in the north and Azogues in the south. Similar rocks at Ambuqui, to the east of Ibarra, near to the Colombian border, are also assigned to this division (Fig. 2A).

Lithologically, the division consists of a continentally derived sequence of orthoquartzites intercalated with low-grade phyllites or slates. The quartzites, which are sometimes feldspathic, vary from medium- to coarse-grained types through to pebble conglomerates; clastic blue quartz is sometimes present.

In the south, the limits of the Guamote division coincide with the Ingapirca fault in the west, and the Peltetec fault in the east (Fig. 2). In this southern area it is noteworthy that a penetrative first cleavage is subparallel to bedding and generally gently dipping, usually to the east. This contrasts strongly with the dominantly vertical structures recorded to the east of the Peltetec fault. Small-scale folds and 'ramps' indicate tectonic transport to the west. Over the entire outcrop, late, 070°-trending, upright, open-to-closed folds, associated in places with a subvertical crenulation

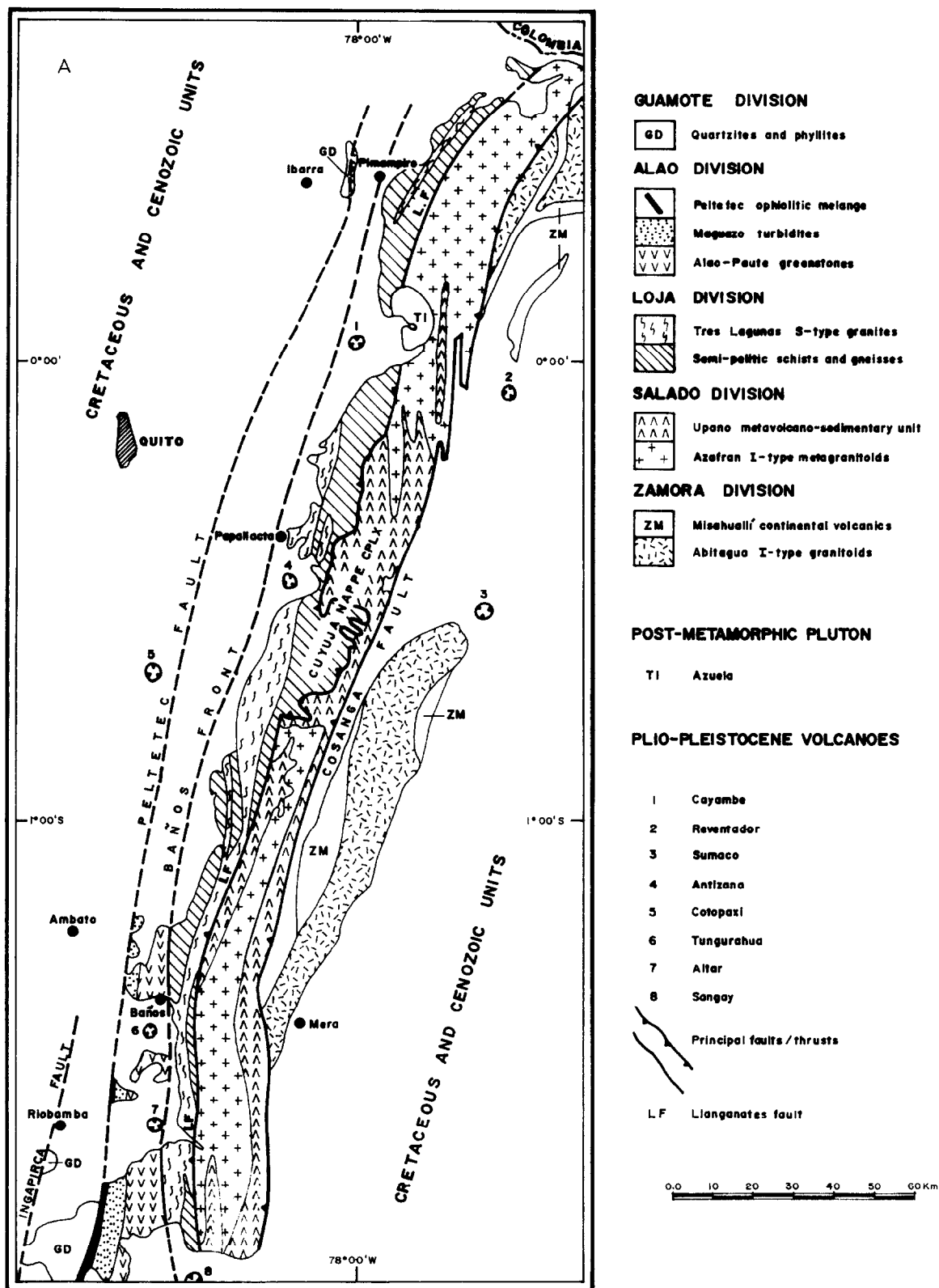
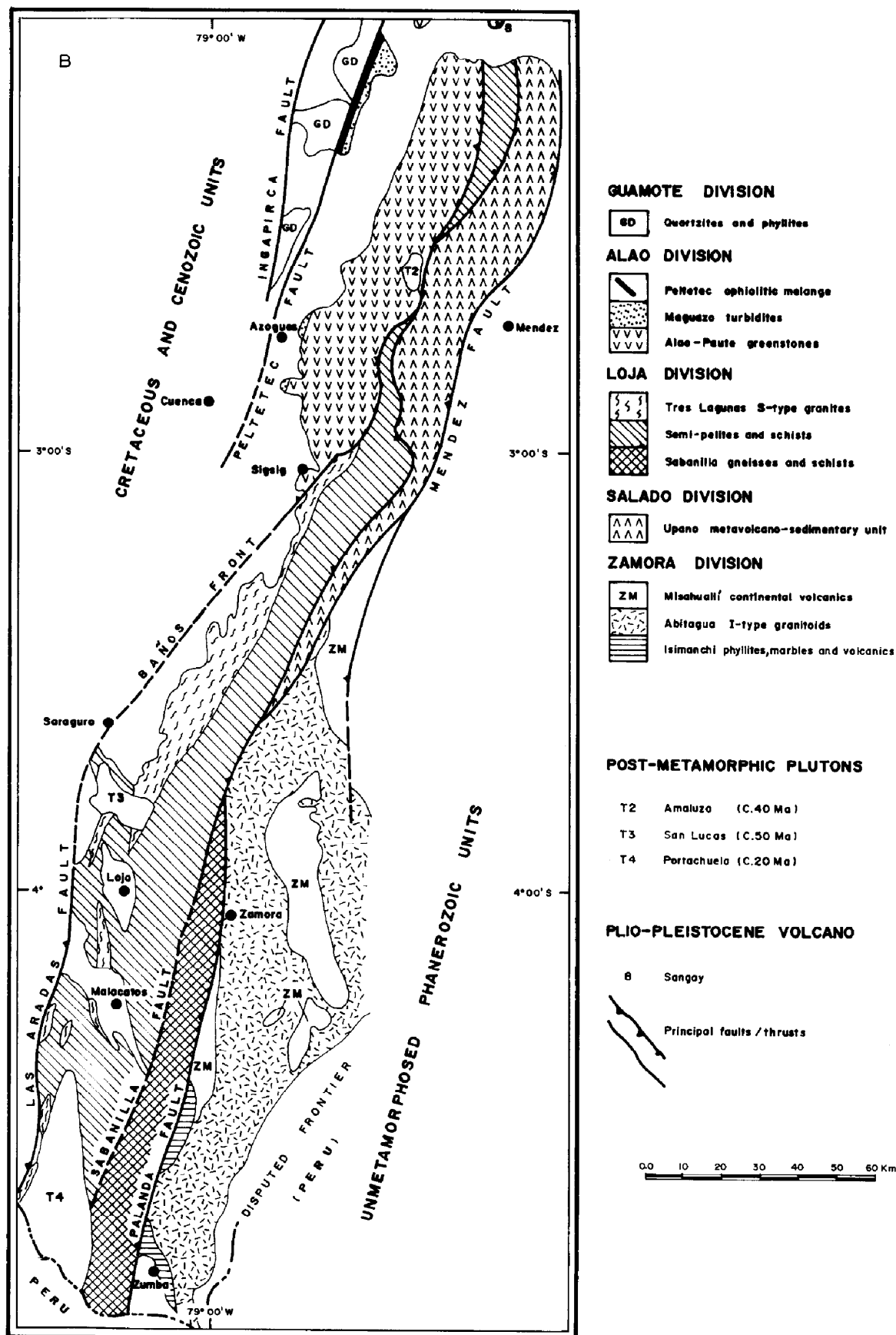


Fig. 2. Simplified geological maps of the pre-Cretaceous rocks of the Cordillera Real and sub-Andean zone north of 2°S (A) and south of 2°S (B).



cleavage, are present. These we relate to younger, possible Cainozoic tectonism.

The age of the Guamote division has not been established directly. In the Riobamba area it is cut by a small, undeformed, hornblende biotite granodiorite stock (Pungala) which has yielded concordant (Hb/Bi) K/Ar ages of 42 ± 1 Ma (Rundle, 1988). The Guamote division is also overlain unconformably by the unmetamorphosed, Maastrichtian Yunguilla Formation of Bristow et al. (1975), which is also affected by the late, upright-folding event referred to above.

Alao division

The Alao division crops out along the western margin of the Cordillera Real, principally to the east of the area between Ambato in the north and Cuenca in the south. Elsewhere, it is assumed to be largely covered by the extensive Plio–Pleistocene volcanic deposits which blanket

much of the Ecuadorian Andes. The structural limits of the division in the east and west coincide with the Baños front and the Peltetec fault, respectively (Fig. 2).

The division is lithologically variable and a number of informal subdivisions have been recognised. In the extreme west, and cropping out along the line of the Peltetec fault, is the Peltetec subdivision interpreted to be an ophiolitic sequence that has been deformed by a series of Andean-trending, subvertical to vertical shear zones. It comprises a series of narrow (< 2 km) outcrops that include cherts and phyllites, spilited basalts, dolerites, serpentinites, gabbros and peridotite (Fortey, 1990). Minor tectonic lenses of Tres Lagunas type (see Loja division) granite also occur.

The Peltetec fault separates 'oceanic' rocks from the continentally derived Guamote division and this same tectonic line can be traced northwards, almost to Ibarra (Fig. 2A), as a neotectonic lineament on satellite imagery.

TABLE 1

Summary of the Pre-Cretaceous geology of the Cordillera Real and sub-Andean zone

DIVISION (west to east)	GUAMOTE		ALAO		LOJA		SALADO		ZAMORA
SUBDIVISION / LITHOLOGIES	Dark and pale ortho- quartzites with slate/ shale bands	PELTETEC FAULT	Peltetec: dismembered ophiolite Maguazo: turbidites Alao-Paute: andesitic greenstones, tuffs and sediments	BAÑOS FRONT	Tres Lagunas: biotite (garnet) granite and orthogneiss Sabanilla: ortho- and paragneiss, Associated with semi- pelitic phyllites, schists and paragneisses	LLANGANATES FAULT	Azafran: calc-alka- line batholith chain (diorite/granodiorite) Upuno: andesitic greenstones, tuffs black phyllites, greywackes and minor marbles	COSANGA - MENDEZ FAULT	Abitagua: calc-alkaline batholith chain Misahualli: andesites, dacites, basalts and agglomerates Isimanchi: marbles and volcano-sedimentary rocks
TECTONO-META- MORPHIC STATE	Very low-grade rocks overthrust W		Low-grade rocks, steep fabrics, upright folds		Low-to medium grade rocks thrust E with imbrications		Low-grade rocks thrust E with im- brications. High - level stornfield and serpentinite klipps		Essentially undeformed and unmetamorphosed
AGE	?Upper Jurassic		Upper Jurassic (Oxfordian-Callovia)		?Triassic plutons in ?Palaeozoic sediments		Jurassic, with possible pre- Jurassic elements		Isimanchi: Triassic Igneous rocks: Jurassic
INTERPRETATION	Continental sediments/ clastic wedge		Ocean floor, forearc and volcanic arc or marginal basin		S-type granites in continentally-derived sediments		I-type plutons in volcano-sedimentary sequence		Continental I-type plu- tonic-volcanic arc

The Peltetec subdivision exhibits an eastern tectonic contact with the Maguazo subdivision, a 5–10 km wide belt that can be traced, albeit in inliers, for c. 200 km between Ambato and Cuenca. Further to the north, to the east of Ibarra (Fig. 2A), the eastern outcrop of what was formerly referred to as the Ambuqui Group (Baldock, 1982) is now included in the Maguazo subdivision.

The Maguazo subdivision is dominated by turbidites, in places rich in volcanic clasts, and andesitic greenstones. Green, metamorphosed tuffs, carbonaceous slates, minor amounts of marbles, orthoquartzites, and cherts are also present. Graded beds indicate that the sequence is right way up and it is folded into a tight-to-isoclinal regional syncline which has a steeply dipping axial plane, and plunges gently to the south.

Further to the east, and at least in part in faulted contact with the Maguazo subdivision, is the extensive Alao–Paute subdivision, outcrops of which are almost continuously exposed between 1°S and 3°S (Fig. 2). These rocks, first described by Sheppard and Bushnell (1933), were previously included in the Paute Series/Group (Bristow, 1973; Baldock, 1982) and consist dominantly of andesitic greenstones and greenschists. In some areas, especially to the northeast of Cuenca (Fig. 2B), metasedimentary rocks, including graphitic phyllites, quartz–silicate and clinozoisite–tremolite rocks, are present (see also Bristow and Guevara, 1980).

In the field it can often be demonstrated that the development of schistosity relates to the presence of generally steep-to-vertical, Andean-trending shear zones and that away from these zones the rocks are often more massive and frequently preserve relict, igneous textures. Generally, the mineralogy is characteristic of greenschist facies with widespread development of chlorite $\pm$ albite $\pm$ quartz $\pm$ epidote $\pm$ biotite $\pm$ actinolite. To the east of Cuenca, volcanic breccias and agglomerates are common and some contain strongly flattened clasts with marked trans-Andean orientation, which suggests that substantial 'in-situ' rotation may have accompanied deformation.

As noted above, the eastern limit of the Alao division corresponds to the Baños front, a re-

gional structure of fundamental importance. First noted to the east of Baños (Fig. 2A), the Baños front corresponds to a change in lithology and, in many places, metamorphic grade, across a varying width of generally steep-to-vertical mylonitic rocks. An exception is the Río Paute section, immediately to the east of the Amaluza pluton (T² in Fig. 2B), where the Alao–Paute greenstones/greenschists are juxtaposed tectonically against similar rocks of the Salado division (see below). The Baños front marks the eastward appearance of the pelitic schists, gneisses and meta-granites of the Loja division.

At Baños (Fig. 2A), foliation is essentially vertical and sigmoidal quartz eyes indicate dextral movement along the front. However, at Sigsig dips are moderate-to-steep towards the west and kinematic indicators suggest the eastwards transport of the Alao division over the Loja division (Fig. 2B). The presence of isolated, tectonic lenses of greenschists along the Las Aradas fault, to the south of Saraguro (Fig. 2B), and the presence of Loja division rocks immediately to the east, strongly suggest that this fault, which marks the western limit of the present-day Cordillera Real in southern Ecuador, represents the continuation southwards of the Baños front. To the north of Ambato (Fig. 2A) the Baños front is tentatively projected under the Cainozoic volcanic cover and assumed to pass close to the small village of Pimampiro (Fig. 2A).

The age of the Alao division is not precisely known but Bristow (1973) considered there to be a transitional contact between these rocks and the volcanic Macuchi Formation and Maastriichtian Yunguilla Formation in the west. However, having re-examined this area, we have found no compelling evidence to support this conclusion and, although more detailed work is required, we interpret the Alao division to be unconformably overlain by the unmetamorphosed Yunguilla Formation.

Various K/Ar determinations have been carried out on the Alao division (e.g., Kennerley, 1980; Rundle, 1988). The ages obtained range from c. 90 to 140 Ma but, without exception, these are considered to be unreliable as primary metamorphic ages due to the altered nature of

the material and, in some cases, the very low K-content obtained from analysed minerals. At present the best estimate for the age of the division is based on palynoflora contained in float samples of the Maguazo subdivision collected to the east of Cuenca (Fig. 2B). These include a variety of Middle/Late Jurassic taxa, in particular *Tubotuberella eisenackii*, which is confined to the Callovian and Oxfordian stages (c. 156–169 Ma) (Riding, 1989).

Loja division

Rocks belonging to the Loja division can be traced along the entire length of the Cordillera Real but they are particularly extensive in the area between Cuenca and the Peruvian border. In the west the division is limited by the Baños front. In the east, to the north of c. 4°S, it is in tectonic contact with, and structurally overlies, the Salado division (Fig. 2). Further to the south it is overthrust along the westerly dipping Palanda fault over the Zamora division (Fig. 2B). To the north of Baños, the principal fault which separates the Loja and the Salado division is the Llanganates fault (Fig. 2A).

Lithologically, the division consists of a variety of rock types but it essentially comprises variably metamorphosed, semi-pelitic rocks and the metagranitoid subdivision of Tres Lagunas. These latter rocks had been previously noted to the east of Saraguro (Kennerley et al., 1973), to the south of Sigsig (Harrington, 1957) and in the Papallacta area (P. Duque, pers. commun., 1986) (Fig. 2) but the present study has confirmed that they occur throughout much of the Cordillera Real. Normally these rocks are strongly foliated and conform to S-C type I mylonites, as defined by Berthé et al. (1979) and Lister and Snoke (1984). They are compositionally restricted and range from biotite ± muscovite granodiorites to monzogranites. In the more massive parts of the intrusions, the Tres Lagunas subdivision is typically medium- to coarse-grained and carries alkali feldspar megacrysts. Hornblende has not been recorded in these rocks but garnet is normally present and, occasionally, cordierite. In addition, many sam-

ples contain crystals of conspicuous, pale-blue quartz, the origin of which is probably related to the presence of microshears that affect the optical properties of crystal lattices.

Xenoliths within the Tres Lagunas subdivision are relatively rare but greenschists, quartzites and 'aplitic' material have been observed. Partially assimilated, semi-pelitic xenoliths and a series of deformed (?syntectonic) amphibolite dykes are present in river blocks to the east of Baños (Fig. 2A).

Based on their mineralogy and K_2O/Na_2O values, these granitoids can be classified as 'S-types' (Chappell and White, 1974) and the suite also has consistently high initial $^{87}Sr/^{86}Sr$ ratios (> 0.712) (Rundle, 1987; Harrison, 1989). Taken together, the above suggests that crustal contamination was an important factor in the genesis of the Tres Lagunas subdivision and it serves to distinguish these rocks from the more typical, 'I-type', plutons of the Ecuadorian Andes.

Hosting the metagranitoids, especially to the north of 2°S (Fig. 2A), are garnet–biotite schists and paragneisses with minor amphibolites. In the south however, low-grade phyllites, quartzites and semi-pelitic schists predominate, but towards the east these are replaced by a narrow elongate belt of medium- to high-grade schists and gneisses of the Sabanilla subdivision (Fig. 2B); a complex unit comprising mainly foliated, possibly syntectonic, in part migmatitic, biotite ± muscovite granitoids. The associated metasedimentary rocks frequently contain garnet, and staurolite. Sillimanite and kyanite have also been recorded (see also Trouw, 1976). Hornblende ± biotite amphibolites are relatively common, especially within the metaplutons, where their form suggests they represent minor intrusions. The origin of the granitoids is enigmatic but, although they lack certain characteristics of the Tres Lagunas subdivision (i.e. absence of alkali feldspar megacrysts and blue quartz), they also have relatively high initial $^{87}Sr/^{86}Sr$ ratios (0.7088 to 0.711) and the available analyses, based on K_2O/a_2O values similarly classify them as 'S-type' granites according to the criteria of Chappell and White (1974).

North of Baños and west of the Llanganates fault, the Loja division rocks are characterised by

a subvertical or steep, west-dipping, Andean-trending, second schistosity. Mineral lineations are horizontal (Andean-trending) or plunge at gentle-to-moderate angles to the south. Narrow belts of flat, tectonic foliation occur but these are essentially monoclinical in form. In the Cuyuja nappe complex (Figs. 2A and 3A) rocks of the Loja division form the middle tectonic level of a subhorizontal belt of nappes that overlie the Salado division and include thin (centimetre to me-

tre scale), tectonic slivers of Tres Lagunas granitoids and isolated lenses of serpentinite.

South of Baños, the Loja division is dominated by an eastwards (tectonic) progression from the Tres Lagunas subdivision, through an extensive semi-pelitic sequence into the Sabanilla subdivision. All these units are cut by Andean-trending shear zones and a D2 tectonic foliation which is generally steeply dipping to the west. Limited belts, characterized by gentle-to-flat (probably

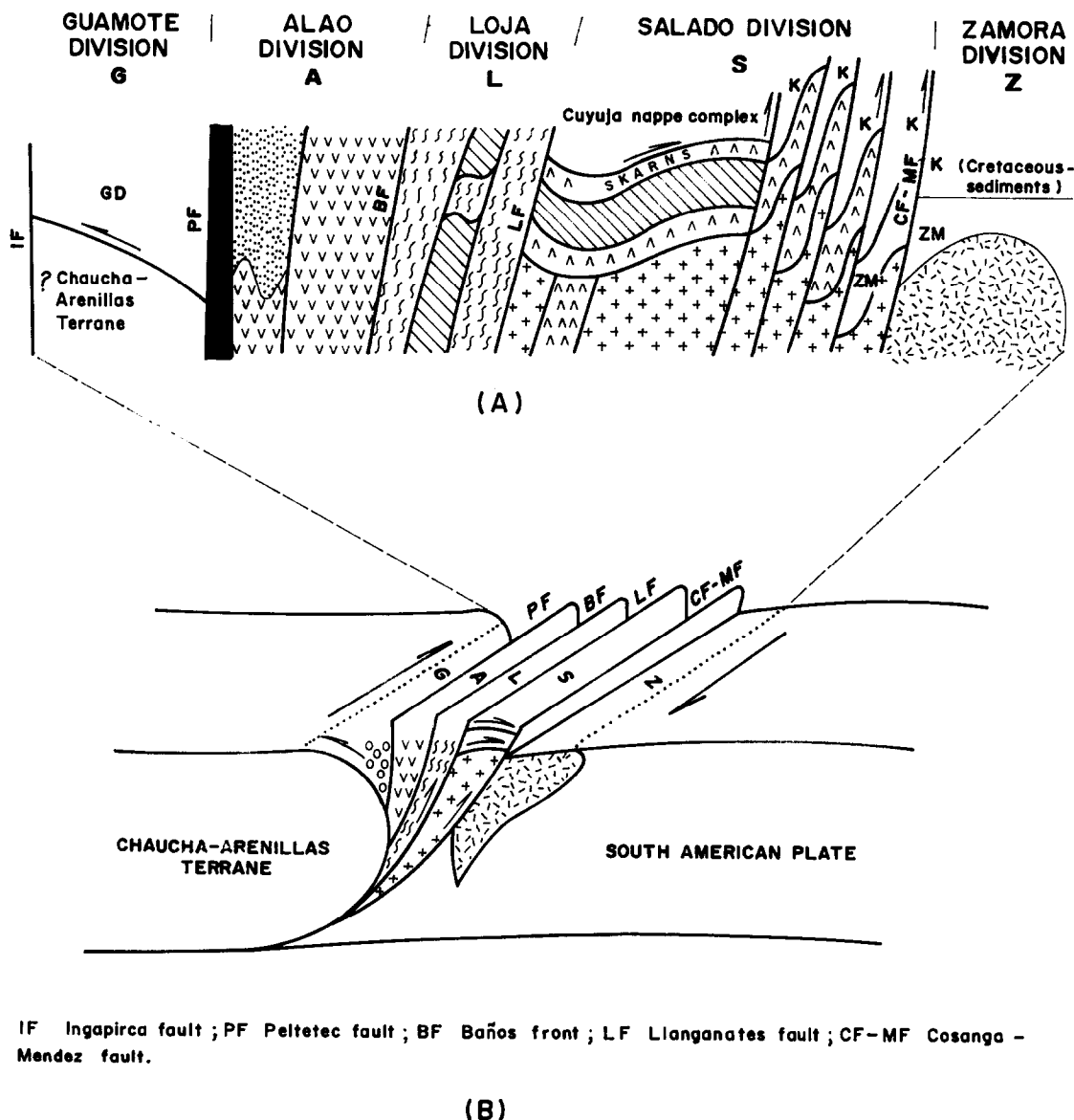


Fig. 3. (A) Schematic section across the Cordillera Real (see Fig. 2 for stratigraphic details); (B) possible collision model to account for the disposition of the individual lithotectonic divisions.

monoclinical) S2 foliation occur, especially in the west, towards the Baños front.

Kennerley (1980) considered the semi-pelitic rocks of the Loja division (part of his Zamora series) to be Palaeozoic in age on the basis of their supposed correlation with similar lithologies in Peru (Gerth, 1955). As yet we have failed to discover sufficiently well-preserved, diagnostic fossils within the low-grade parts of the Loja division, nor have our attempts to date the higher-grade units and granitoids radiometrically been entirely successful. A single Sm–Nd (garnet) isochron from the Tres Lagunas granite to the east of Saraguro (Fig. 2B) gave an age of 257 ± 125 Ma (Harrison, 1989). Whole-rock, Rb–Sr 'errorchrons' gave ages of 194 ± 50 Ma (MSWD 49.5) and 189 ± 43 Ma (MSWD 289.1) and a combined (18 point) 'errorchron' gave 200 ± 12 Ma (MSWD 169.1) (Harrison, 1989). The following 'errorchron' ages (Rb–Sr, whole-rock) have also been obtained from various orthogneisses within the Sabanilla subdivision: 198 ± 45 Ma (MSWD 35); 233 ± 51 Ma (MSWD 175); 234 ± 19 Ma (MSWD 206); and 224 ± 37 Ma (MSWD 108) (Rundle, 1988; Harrison, 1989). Based on the above we conclude that the best estimate for the minimum age of the granitoids of the Loja division is probably somewhere between 200 and 220 Ma.

More than 40 K/Ar mineral determinations have also been carried out on various samples from the Sabanilla and Tres Lagunas subdivisions. These dates, considered to be disturbed ages, range from 105 to 45 Ma with a marked peak between 85 and 65 Ma (Aspden, 1990). Samples from the higher-grade envelope rocks of the Tres Lagunas subdivision, near Papallacta (Fig. 2A), have yielded older K/Ar ages of 324 ± 16.5 Ma (Hb), 367 ± 9.5 Ma (Hb) and 863 ± 32 Ma (Bi) (Rundle, 1987; Harrison, 1989) and suggest the presence of an older basement. Further work is required in order to test this possibility.

Salado division

The Salado division is especially widespread to the north of 3°S, but to the south it is eliminated tectonically and probably stratigraphically. In the

north, its eastern limit, which appears to be transitional with the largely undeformed Zamora division, coincides regionally with the Cosanga and Mendez faults. These faults are considered to represent the western limit of the cratonic front which, at depth, is assumed to approximate to the western edge of the Precambrian Amazonian craton.

Two principal subdivisions, the plutonic Azafran and the volcano-sedimentary Upano, are recognised within the Salado division.

Although previous work along the Mera road, to the east of Baños (Fig. 2A), had recognised the presence of the variably deformed Azafran granite (Sauer, 1958; Kennerley, 1971; Mortimer et al., 1980), this pluton was considered to be an isolated body of limited extent. The present study, however, has shown that the pluton in fact represents only a small part of a batholithic chain which can be traced for almost 300 km, from the Colombian border in the north to c. 2°S (Fig. 2A). In the north, the Azafran subdivision is represented by the Chingual and Sacha plutons which typically comprise variably deformed and gneissic, coarse- to medium-grained biotite $\pm$ hornblende granodiorites and tonalites. Subordinate diorites, hornblendites and gabbros are also present, and both deformed and undeformed mafic (hornblende and/or biotite-rich) xenoliths are common. To the south, identical rocks have been encountered on various foot traverses across the Cordillera but they are absent along the main road to the east of Papallacta, where they are assumed to be covered by the Cuyuja nappe complex (Figs. 2A and 3A). Along the Baños road, the limits of the Azafran granite (see Mortimer et al., 1980) have also been extended westwards to the Llanganates fault (Fig. 2A) to include a variable sequence of orthogneisses, schists and hornblende diorites.

The Upano subdivision is a mixed volcano-sedimentary sequence which includes metamorphosed andesites, tuffs and agglomerates, greywackes, marbles, impure quartzites and black phyllites. The marble sequence of Cerro Hermoso (Sauer, 1958) is over 500 m thick (Litherland et al., 1990). As is common elsewhere in the Cordillera, these rocks are variably deformed and,

although metamorphism is generally within the greenschist facies, hornblende amphibolites are occasionally present. In the more pelitic horizons of the Upano subdivision, muscovite, biotite, garnet and chloritoid are common and kyanite is also locally developed (Litherland et al., 1990).

To the north of Baños, a series of isolated, high-level tectonic klippe of skarn are present between the Llanganates and the Cosanga faults (Fig. 3A). Although erosion has now removed these rocks except at the highest level, they are preserved extensively within the Cuyuja nappe complex, and can be traced, discontinuously for at least 150 km along the Cordillera. The skarns, which in some areas are also associated with thin sheets of serpentinite, are of the calcic magnetite type (Einaudi et al., 1981), and are considered to have been formed from an Upano subdivision protolith, representatives of which are found at the base of the nappe complex and below the roof thrust (Fig. 3A). The model proposed by Litherland et al. (1990) envisages the Azafran plutonic phase and the Llanganates fault to be essentially contemporaneous with the Upano subdivision volcanic and sedimentary rocks thrust eastwards over the hot pluton, to form the high levels of the Cuyuja nappe complex.

It is apparent that various tectonic regimes are present within the Salado division (Fig. 3A). The Azafran subdivision, although not uniformly deformed, almost everywhere exhibits a vertical to generally steep, westerly-dipping foliation, which can often be related to the presence of Andean-trending shear zones. In several places along the road section to the east of Baños, weakly foliated to massive plutonic rock can be seen to pass through variably foliated orthogneiss into a schistose variant that normally marks the central portion of the shear zones where deformation was most intense. As was the case in the Tres Lagunas subdivision, S-C type I mylonites (Lister and Snoke, 1984) are widely developed. Mineral lineations, though locally steep, normally have gentle ($< 30^\circ$) Andean plunges or are subhorizontal. Preliminary kinematic studies of S-C fabrics indicate that dextral movements were dominant.

The Cuyuja nappe complex (Fig. 2A) structurally lies some 3 km above the level of the

Baños road section, and contains rocks from both the Salado and Loja divisions. Within it, subhorizontal, eastward-directed thrust sheets are present above the steeply foliated Azafran subdivision. Interestingly, in this area, mineral lineations are also Andean-trending suggesting an oblique (transpressional) control.

Towards the sub-Andean zone, near the Cosanga and Mendez faults, the Upano division is in tectonic contact with the Zamora division. This zone is considered to have been active throughout the Mesozoic. However, it has also been affected by Tertiary thrusting, principally Late Miocene to Early Pliocene (Kennerley, 1980; Baldock, 1982), which in places has brought the older greenstone/greenschist units of the Upano subdivision into tectonic contact with the Cretaceous sediments of the Hollín, Napo and Tena Formations (Fig. 3A).

An eight-point, whole-rock Rb/Sr isochron from the foliated Chingual pluton, located near the Colombian border, gave an age of 156 ± 21 Ma (MSWD 2.8) and a similar seven-point isochron from the Azafran 'granite' to the east of Baños, gave an age of 120 ± 5 Ma (MSWD 2.4) (Rundle, 1987). Two samples of almost identical hornblende-biotite diorite, collected to the west of the Azafran 'granite', gave the following concordant K/Ar mineral ages: (A) 175 ± 5 Ma (Hb), 175 ± 5 Ma (Bi); and (B) 128 ± 4 Ma (Hb), 125 ± 4 Ma (Bi) (Rundle, 1988). These samples were collected only a few metres apart; however, (B) is from the margins of a shear zone, whereas (A) comes from a completely massive and apparently unaffected part of the pluton. We interpret the younger ages to be reset by the shearing event and suggest that the older dates possibly represent original magmatic cooling ages which, although somewhat older, are not dissimilar to the date of 156 Ma obtained from the foliated Chingual pluton. If one accepts this interpretation then the status of 120 Ma isochron age obtained from the Azafran 'granite' is brought into question. Our current interpretation is that this has also probably been reset during the regional shearing event (i.e. c. 120–130 Ma), but zircon analysis planned for the future will hopefully resolve this problem.

No reliable age determinations or palaeontological control exists for the Upano subdivision but it is tentatively considered to represent the largely contemporaneous (i.e. Middle to Late Jurassic) volcano-sedimentary envelope of the Azafran pluton chain and to be transitional with the Jurassic Misahualli subdivision further east. It should be noted however, that the presence of older elements can not be ruled out.

Zamora division

The Zamora division occurs immediately to the east of the Cordillera Real proper, close to what is considered to be the approximate western edge of the Amazonian craton. The Zamora division comprises two principal subdivisions, the plutonic Abitagua and the volcanic Misahualli, which are considered to be broadly contemporaneous and the age equivalents of the Salado division. It also includes the poorly known Isimanchi subdivision in the southeastern part of the Cordillera Real (Fig. 2B). The change from the Misahualli volcanic sequence, which is mainly continental, to the marine volcano-sedimentary Upano division, takes place across the Cosanga fault (Fig. 2A). Further to the south, the western limit of the non-foliated Zamora division is defined by the Mendez and Palanda faults. Together these three faults also mark a natural cratonic limit which, with the exception of the Isimanchi subdivision (see below), separates metamorphosed rocks in the west from unmetamorphosed rocks in the east.

The Abitagua subdivision consists of three, essentially undeformed, calc-alkaline batholiths. From north to south these are the Rosa Florida, Abitagua and Zamora batholith (Fig. 2), the latter of which now includes the Río Mayo batholith, near to the Peruvian border, which was originally thought to represent a separate and younger intrusion (Baldock, 1982).

In the north, the Misahualli subdivision consists of agglomerates and green tuffs intruded by subvolcanic and plutonic rocks of the Rosa Florida pluton which vary from quartz syenite to quartz monzonite in composition. Similar green and purple tuffs, lavas and agglomerates are pre-

sent in the area to the west of the Cosanga fault, where they are deformed and contain sedimentary units similar to those of the marine Upano subdivision.

The Abitagua batholith intrudes undeformed, porphyritic, silicic lavas, associated flow breccias and pyroclastic rocks. Further south, feldspar microporphyritic andesites, hornblende andesites, and dacites are associated with the Zamora batholith as are a series of small, high-level, subvolcanic intrusions. Some of these latter intrusions are associated with polymetallic gold mineralisation and it is probable that they relate to a younger (post-batholith) phase of activity.

Prior to this study, the age of the Abitagua and Zamora batholiths was only poorly constrained. Kennerley (1980) reported K/Ar mineral ages of 152 ± 4 Ma (Kspar), 173 ± 5 Ma (Hb) and 180 ± 5 Ma (Bi) from a single sample from the Zamora batholith, and Pichler and Aly (1983) also obtained a K/Ar date of 171 ± 6 Ma (Bi). A three-point, whole-rock, Rb/Sr isochron age of 173 ± 5 Ma was obtained by Halpern (quoted in Hall and Calle, 1982) for the Abitagua batholith; Herbert (1977) gives a K/Ar (Bi) age of 178 ± 7 Ma and a slightly older K/Ar (Bi) age of 194 ± 7 Ma was reported by Pichler and Aly (1983).

During the past four years the project has dated various plutonic rocks from both the Abitagua and Zamora batholiths. Samples from Abitagua gave two separate Rb/Sr whole-rock isochrons with ages of 161 ± 2 Ma (MSWD 0.9) and 163 ± 2 Ma (MSWD 2.5). The combined results from these samples produced a 16-point isochron and an age of 162 ± 1 Ma (MSWD 2.5) (Rundle, 1987). Rb/Sr results obtained from the Zamora batholith failed to define an isochron, but over 20 K/Ar determinations have been obtained (Rundle, 1988, 1990; Harrison, 1989) and several samples have yielded concordant hornblende/biotite mineral ages. Since the Zamora batholith is undeformed, these dates are taken to represent magmatic cooling ages and they indicate that plutonism ranged from c. 150 to 190 Ma.

The age of the Misahualli subdivision is not well-established but we assume it to have a similar age range to the Abitagua subdivision. A

single K/Ar (Hb) date of 230 Ma (Rundle, 1988) may indicate the existence of older material.

In the extreme southeast of the Cordillera Real is a distinctive, but relatively poorly known, mixed suite of low-grade metamorphic rocks, the Isimanchi subdivision, which, for convenience, are also included within the Zamora division. In the west these rocks are in tectonic contact with, and overthrust by, the Sabanilla subdivision along the Palanda fault (Fig. 2B). In the east they are intruded by, and occur as large, kilometre size, roof pendants within, the Zamora batholith.

Lithologically the unit consists of a metamorphosed, immature, volcano-sedimentary sequence comprising phyllites, dark-coloured, fine-grained (?) tuffs, poorly sorted siltstones, rich in volcanic debris, and prominent marbles. It is possible that the Isimanchi division represents the protolith of the important gold-bearing, grandite skarns of the Nambija area, located within the Zamora batholith and situated c. 20 km due east of Zamora (Fig. 2B).

The age of this division is not well-established. However, bivalves recovered from a large xenolith of the presumed Isimanchi subdivision within the Zamora batholith are of late-Middle to Late Triassic, probably Norian, age (Ivimey-Cook and Morris, 1989).

Other pre-Abitagua subdivision rocks of the sub-Andean zone

In addition to the Isimanchi subdivision, various other pre-Abitagua subdivision, but essentially unmetamorphosed, units are also present in the sub-Andean zone. Although some of these have not specifically been studied during the current project, their presence is, nevertheless, important in terms of the regional geology.

The Zumba mafic-ultramafic complex, located near to the town of the same name close to the Peruvian border (Fig. 2B), includes serpentinites, quartz gabbros and hornfelsed orthopyroxene norite. Immediately to the east, xenoliths of hypersthene gabbro and strongly chloritised and epidotised rocks (Fortey, 1990), interpreted to be related to the Zumba complex, are present within the Zamora batholith. Elsewhere in the sub-

Andean area, contact-metamorphosed basaltic pillow lavas and hyaloclastites have recently been discovered along the eastern margin of the Zamora batholith (I. Gemuts, pers. commun., 1990) and further to the north at Mendez (Fig. 2B), isolated outcrops of tholeiitic pillow basalts are known (F. Van Thournout, pers. commun., 1990). To the east of Mendez, along the western flanks of the Cutucu uplift (Baldock, 1982), basaltic lavas, in places with pillows, are exposed along new road cuts of the projected trans-Amazon highway. Our observations indicate that these rocks occur within an extensive, continental-type sequence of tuffaceous grey siltstones and sandstones which can be traced laterally (eastwards) into the turbiditic Santiago Formation (see also Tschopp, 1953). Ammonites recovered from the Santiago Formation (Tschopp, 1953; Geyer, 1974; Ivimey-Cook, 1989) indicate a Sinemurian age (c. 200–206 Ma) and the Santiago Formation can thus be correlated with similar rocks in northern Peru: the Aramachay Formation of the Pucara Group (Megard, 1968; Jaillard et al., 1990) where, as noted by Baldock (1982), there is an equivalent facies change between the marine Pucara Group in the east and the volcano-clastic Zaña Group in the west (Cobbing et al., 1981).

Northwards of c. 2°S, the marine Santiago Formation is absent but, according to Tschopp (1953), along the eastern margin of the Cutucu uplift, it is overlain unconformably by a succession of continental redbeds, the Chapiza Formation. These rocks, however, are lithologically similar to those which occur along the western flank of the uplift and we suggest that it is possible that at least part of the poorly dated Chapiza Formation could be a lateral facies equivalent of the Santiago Formation. The linear form of the Santiago outcrop (see Baldock, 1982), the eastwards transition from volcanic-rich, continental-type deposits in the west, and the presence of basaltic, in some cases tholeiitic, pillow lavas suggest that deposition of the Santiago Formation took place in an elongate, north-northeast–south-southwest trending, extensional basin and that it was associated with widespread volcanic activity, especially along its flanks.

Cretaceous units

In the Oriente and sub-Andean zone (Fig. 1), Cretaceous units comprise the epicontinental quartzites of the Aptian–Albian Hollin Formation which were derived from the east and are conformably overlain by the marine shales and limestones of the middle Albian to lower Campanian Napo Formation (Tschopp, 1953; Bristow and Hoffstetter, 1977; Baldock, 1982). A marked erosional unconformity separates the Napo Formation from the sandstones of the overlying Maastrichtian to possibly lower Campanian Tena Formation (Tschopp, 1953; Bristow and Hoffstetter, 1977; Baldock, 1982), which was derived from the west (Baldock, 1982). In the west of the Cordillera Real there are outcrops of the Maastrichtian, flysch-like, Yunguilla Formation near Cuenca (Bristow, 1973). There are also granodioritic plutons and Alaskan-type mafic/ultramafic pipes (Litherland et al., 1990) of Late Cretaceous age (Harrison, 1989) in the vicinity.

Prior to the deposition of the Hollin quartzites, the pre-Cretaceous basement rocks in the sub-Andean zone were deformed and underwent erosion. Along the Cosanga and Mendez faults, the Cretaceous units are involved in a Late Tertiary ('Andean'), imbricate thrust belt, which also affects Miocene units (Fig. 3A) (Baldock, 1982). It is of interest to note that, within the Cordillera itself, although a large amount of vertical uplift took place at this time, deformation was apparently restricted as evidenced by the presence of a number of undeformed, post-metamorphic, Tertiary intrusions that range from c. 20 to 60 Ma in age (see Fig. 2).

Geological summary and conclusions

The present study has established a preliminary, regional lithotectonic framework for the Cordillera Real in Ecuador which, hopefully, will provide the basis for further work. Major uncertainties remain to be resolved but, nevertheless, sufficient information is available to allow some speculation about the geological history and development of this part of the Northern Andes.

The Amazonian craton in the east was stabilised in the Proterozoic (Litherland et al., 1985) and, during the Palaeozoic, was the site for the accumulation of platform deposits, the Pumbuíza and Macuma Formations (Tschopp, 1953; Baldock, 1982). During the Early Mesozoic, in the sub-Andean zone, a narrow extensional basin began to form along the western edge of the craton, the early stages possibly being marked by the marbles and immature volcano-sedimentary sequence of the (?)Norian Isimanchi subdivision. During Sinemurian time, marine conditions extended as far north as 2°S and led to the deposition of the Santiago Formation. Correlation with similar rocks in northern Peru (Jaillard et al., 1990) suggests that the 'Santiago trough' propagated from south to north and in Ecuador it was flanked in the west (and possibly in the east) by laterally equivalent, volcanic-rich, continental deposits.

At c. 190 Ma major, calc-alkaline, volcano-plutonic activity commenced (the Abitagua and Misahualli subdivisions) and continued until c. 150 Ma. In southern Ecuador it appears that the main plutonic axis coincided with that of the Santiago trough. This same plutonic activity can also be traced northwards into Colombia (Aspden et al., 1987b) and, hence, is of regional significance since it affected the entire Northern Andes. In Ecuador, especially in the north, the Zamora division is paralleled by, and possibly transitional with, the Salado division to the west. The Cosanga/Mendez faults mark the limit of these two divisions and also the change from the essentially 'continental', volcanic sequences of the Misahualli subdivision into the marine, volcanoclastic, Upano subdivision, suggesting that this line was tectonically active during the Middle to Late Jurassic, possibly in the form of a listric fault.

Further to the west is the Loja division, the western limit of which corresponds to the Baños front. The oldest dates recorded anywhere in the Cordillera Real (i.e. pre-Mesozoic) are from this division, but more detailed studies are required before these can be commented on further. Immediately to the east of the Baños front, the Loja division is characterised by the presence of a belt

of 'S-type' plutons (Tres Lagunas subdivision) which extend throughout the length of the Cordillera Real. Such rocks have not previously been recorded in the Northern Andes and, although poorly dated, the best estimate for their age is c. 200–220 Ma.

Preliminary studies in the El Oro province in southwest Ecuador obtained a 10-point Sm–Nd (garnet) isochron age of 219 ± 22 Ma (MSWD 0.4) (Harrison, 1989) from garnet-bearing paragneisses which crop out immediately to the south of the Raspas fault (Fig. 1). This date confirms the existence of a regional metamorphic event during the Late Triassic and it could also hint at a genetic link between the allochthonous metamorphic rocks of El Oro and the Loja division in the Cordillera Real. It has been suggested previously by Aspdén et al. (1988) that the Tres Lagunas granites could relate to the accretion of the gneissic Chaucha–Arenillas terrane along the Peltetec 'suture' but, the present geochronological framework appears to preclude this possibility. Recently, Jaillard et al. (1990) proposed that the Mesozoic evolution of the Northern Andes could be considered in terms of a Tethyan rifting model which, in western Gondwana, began in Late Triassic time. Such a model could explain the presence of extensional regimes, evidence for which is preserved in the sedimentary record of Colombia, Ecuador and northern Peru (Jaillard et al., 1990), and it could also account for the generation of the Tres Lagunas granites. In this scenario, the Baños front would be interpreted to represent the remnants of the encratonic shear zone along which, what is now, the northwestern portion of the South American continental plate separated from the southern part of the North American continental plate.

Along the western margin of the Cordillera Real, limited in the east by the Baños front and in the west by the Peltetec fault, is the Alao division. Immediately to the west of the Baños front this comprises a massive sequence of meta-andesites (Alao–Paute subdivision), but at present we are unable to say whether these rocks formed in an oceanic or marginal basin setting.

The presence of an ophiolitic assemblage, which apparently includes a pelagic cover se-

quence, and is associated with volcanic-rich turbidites in the west (i.e. the Peltetec and Maguazo subdivisions), but the absence of equivalent lithologies to the east of the Alao–Paute meta-andesites would be consistent with the interpretation that the Peltetec fault represents a palaeo-subduction zone. In this context it is also of interest to note that in El Oro, along the Raspas fault (Fig. 1), is the Raspas blueschist complex (Feininger, 1980) from which a single K/Ar (phengite) age of 132 ± 5 Ma has been obtained (Feininger and Silberman, 1982). It is therefore tempting to equate this complex with the ophiolitic Peltetec subdivision, but more detailed studies are required in order to substantiate this.

Although only poorly dated, the recognition of Callovian/Oxfordian taxa (c. 170, 155 Ma) in the Maguazo subdivision (Riding, 1989) suggests that the Alao division is, at least in part, contemporaneous with the plutonic Abitagua subdivision in the sub-Andean zone. If this correlation is accepted then it is not easy to envisage a simple, subduction zone model which could satisfactorily explain the present-day relative positions of these two units.

To the west of the Peltetec fault lies the continentally derived Guamote division. As mentioned earlier, the Chaucha–Arenillas terrane is considered to be present at depth in this area and it is envisaged that during the Mesozoic this gneissic terrane largely sourced the Guamote division as it approached from the west/southwest during the closure of the Alao ocean/marginal basin. This closure, took place along the Peltetec fault following cessation of volcano-plutonic activity in the Zamora division (i.e. c. 150 Ma), but prior to the deposition of the Hollin quartzite in the east. During this period, the Guamote division was thrust to the west while to the east of the Peltetec line tectonic transport was to the east (Fig. 3). It is probable that the Peltetec collision was oblique (transpressional) since this would explain both the major overthrusts (e.g., Cuyuja nappe complex, Figs. 2A and 3B) and the essentially north-south, dextral movements deduced along the steep-to-vertical, Andean-trending shear zones. The common occurrence of S-C type mylonites in the Cordillera Real suggests that transpressional

movements have been of fundamental importance in shaping the tectono/structural development of the Cordillera Real. Evidence from the Azafran subdivision, quoted earlier, is interpreted to indicate that major shear zones within the Cordillera Real were (?still) active at c. 125 Ma. Figure 3B shows a schematic section through the Cordillera Real illustrating the main elements of this latest Jurassic to middle Early Cretaceous collisional event. Two noteworthy features which could help in the interpretation of this event are: the presence of blue quartz (?from the Tres Lagunas granite) in the Guamote sediments, and the presence of tectonic lenses of Tres Lagunas granite within the Peltetec subdivision.

As a result of this collision, the pre-Cretaceous rocks in the Cordillera were deformed and metamorphosed (often dynamically). To the east of the Cosanga–Mendez fault (i.e. the cratonic front) regional metamorphism is lacking, but folding, uplift and erosion took place prior to the deposition of the Hollin Formation which everywhere rests with marked unconformity on pre-Cretaceous units. Unfortunately, the base of the Hollin Formation is not precisely dated (Bristow and Hoffstetter, 1977), but from c. 120 Ma (i.e. the base of the Aptian), conditions must have been relatively stable as the epicontinental Hollin quartzites were laid down from the east in an extensive shelf environment (Baldock, 1982). Similar conditions of relative stability also probably existed during the deposition of the marine Napo Formation (c. 110–83 Ma).

In the Cordillera, a major thermal event occurred sometime between c. 85–55 Ma and resulted in a widespread disturbance of isotope systematics. Numerous K–Ar dates, especially from the pre-Cretaceous Sabanilla and Tres Lagunas subdivisions, give ages within this range, but with a marked peak between 85 and 65 Ma (Aspden, 1990). Such dates led Feininger (1982) to propose that the principal metamorphic event in the Cordillera Real was Late Cretaceous in age, but we regard this as a resetting event which affected not only the Cordillera Real in Ecuador but also the Central Cordillera in Colombia (McCourt et al., 1984). Regionally, this event may correspond to the approach and subsequent ac-

cretion of the allochthonous, oceanic Western Cordillera along the Calacali–Pallatanga–Palenque fault (suture) (Fig. 1) in Ecuador and, along its northern equivalent, the Cauca–Patía fault in Colombia (McCourt et al., 1984; Aspden et al., 1987a). In eastern Ecuador, erosion of the top of the Napo Formation occurred between c. 83 and 73 Ma prior to the deposition of the overlying Maastrichtian–?Lower Palaeocene (c. 73–?60 Ma) redbed Tena Formation (Baldock, 1982). At the same time in the west, the marine (Maastrichtian) Yunguilla Formation was deposited (Bristow, 1973). Together, these events coincide with the peak of reset mineral ages from the Cordillera Real. Sedimentological evidence from the Tena Formation (Baldock, 1982) indicates a sedimentary source in the west and, since this formation is confined to the eastern flank of the Cordillera Real, it seems reasonable to conclude that the Late Cretaceous–?earliest Tertiary thermal resetting was synchronous with the uplift and the emergence of the Cordillera Real as a positive topographic feature. In spite of the fact that a thermal event affected much of the Cordillera, its regional effect on the metamorphic mineral assemblages and its tectonic imprint within the older metamorphic rocks has yet to be clearly defined. A possible explanation would be to assume that the accretion of the Western Cordillera also took place from the southwest as has been widely suggested (McCourt et al., 1984; Megard, 1987; Daly, 1989). Thus the kinematic framework for both the latest Jurassic–middle Early Cretaceous and the Late Cretaceous–earliest Tertiary collisions would have been similar and would have resulted in the overprinting of older structures by younger, but essentially parallel ones. Such fault rejuvenation can in fact be demonstrated up to recent times. For example, the Peltetec fault at present defines the eastern limit of the inter-Andean graben and shows neotectonic downthrow to the west of Upper Cainozoic volcanics against metamorphic basement. Equally, the sub-Andean fault/thrust system culminated in the Upper Cainozoic. Thus the major faults of the Cordillera Real have long and complex Mesozoic–Cainozoic histories involving strike-slip, thrust and normal movements.

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